MATH 291

Honors Multivariable Calculus

Jeffrey Lu

Contents

1 Lines and Planes **3**

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Theorem 1. Let $V \subset \mathbb{R}^n$, and let $W = Span(V)$. Then $Span(W) = W$

Theorem 2. (Fundamental property of the standard basis). Let $\mathbf{e_1}, ..., \mathbf{e_n}$ be the standard basis in $\mathbb{R}^n$. $Span(\mathbf{e_1}, ..., \mathbf{e_n}) = \mathbb{R}^n$, and for each $\mathbf{x} \in \mathbb{R}^n$, there is exactly one vector of coefficients $(t_1, ..., t_n)$ such that

$$\mathbf{x} = \sum_{j=1}^{n} t_j \mathbf{e}_j$$

namely $(t_1, ..., t_n) = x_1, ..., x_n)$.

Theorem 3. (Cauchy–Schwartz inequality). For any two vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$,

$$|\mathbf{a} \cdot \mathbf{b}| \leq \|\mathbf{a}\| \, \|\mathbf{b}\|$$

There is a inequality iff $\|\mathbf{b}\| \, \mathbf{a} = \pm \|\mathbf{a}\| \, \mathbf{b}$

Theorem 4. (Triangle inequality). For any three vectors $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$,

$$\|\mathbf{x} - \mathbf{z}\| \leq \|\mathbf{x} - \mathbf{z}\| + \|\mathbf{y} - \mathbf{z}\|$$

Theorem 5. The Euclidean distance function

$$\varrho_E(x, y) := \sqrt{(\mathbf{x} - \mathbf{y}) \cdot (\mathbf{x} - \mathbf{y}} = \|\mathbf{x} - \mathbf{y}\|$$

is a metric in $\mathbb{R}^n$

Theorem 6. (Fundamental Theorem on Orthonormal Sets in $\mathbb{R}^n$). Let $\{\mathbf{u_1}, ..., \mathbf{u_n}\}$ be any orthonormal set in $\mathbb{R}^n$ consisting of exactly n vectors. Then every vector $\mathbf{x} \in \mathbb{R}^n$ can be written as a linear combination of the vectors $\{\mathbf{u_1}, ..., \mathbf{u_n}\}$ in exactly one way, namely

$$\mathbf{x} = \sum_{j=1}^{n} (\mathbf{x} \cdot \mathbf{u_j}) \mathbf{u_j}$$

The squared length of $\mathbf{x}$ is the sum of the squares of the coefficients in the expansion:

$$\|\mathbf{x}\| = \sum_{j=1}^{n} (\mathbf{x} \cdot \mathbf{u_j})^2$$

The standard basis of $\mathbb{R}^n, \{\mathbf{e_1}, ..., \mathbf{e_n}\}$ is a set of n orthonormal vectors in $\mathbb{R}^n$, and so the theorem says that

$$\mathbf{x} = \sum_{j=1}^{n} x_j \mathbf{e_j} = \sum_{j-1}^{n} (\mathbf{x} \cdot \mathbf{e_j}) \mathbf{e_j}$$

is the unique way to express any $\mathbf{x}$ in $\mathbb{R}^n$ as a linear combination of the standard basis vectors.

Definition 1. (Linear Combinations and Span). Let $V \subset \mathbb{R}$. The span of V is the set of all possible linear combinations $\sum_{j=1}^{m} t_j \mathbf{v_j}$.

Theorem 7. (Algebraic properties of the cross product). Let $\mathbf{a}, \mathbf{b}$, and $\mathbf{c}$ be any three vectors in $\mathbb{R}^3$, and let t be any number. Then

$$\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$$
$$\mathbf{a} \times (\mathbf{a} + \mathbf{b}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$$
$$(t\mathbf{a}) \times \mathbf{b} = t(\mathbf{a} \times \mathbf{b}) = \mathbf{a} \times (t\mathbf{b})$$

Theorem 8. (Triple product identity). Let $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^n$. Then

$$(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = (\mathbf{b} \times \mathbf{c}) \cdot \mathbf{a}$$

In other words, the triple product $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}$ is unchanged when the vectors in it are cyclicly permuted.

Theorem 9. (Cross product equals normal vector) Let $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$. Then $\mathbf{a} \times \mathbf{b}$ is orthogonal to both $\mathbf{a}$ and $\mathbf{b}$.

Theorem 10. Let $\{\mathbf{u_1}, \mathbf{u_2}, \mathbf{u_3}\}$ be any orthonormal basis of $\mathbb{R}^3$. Then

$$\mathbf{u_1} \times \mathbf{u_2} = \mathbf{u_3} \iff \mathbf{u_2} \times \mathbf{u_3} = \mathbf{u_3} \iff \mathbf{u_3} \times \mathbf{u_1} = \mathbf{u_2}$$

In particular, $\{\mathbf{u_1}, \mathbf{u_2}, \mathbf{u_3}\}$ is right handed if and only if any one of the identities in equation above is valid, and in that case, all of them are valid.

Theorem 11. (Invariance under change of basis). Let $\{\mathbf{u_1}, \mathbf{u_2}, \mathbf{u_3}\}$ be any right handed orthonormal basis in $\mathbb{R}^3$. For any vectors $\mathbf{x} = (x_1, x_2, x_3)$ and $\mathbf{y} = (y_1, y_2, y_3)$, define $\mathbf{a} = x_1\mathbf{u_1} + x_2\mathbf{u_2} + x_3\mathbf{u_3}$and $\mathbf{b} = y_1\mathbf{u_1} + y_2\mathbf{u_2} + y_3\mathbf{u_3}$, so that $\mathbf{x}$ and $\mathbf{y}$ are the coordinate vectors of $\mathbf{a}$ and $\mathbf{b}$ with respect to the coordinate system based on the orthonormal basis $\{\mathbf{u_1}, \mathbf{u_2}, \mathbf{u_3}\}$. Then

$$\mathbf{a} \times \mathbf{b} = (x_2y_3 - x_3y_2)\mathbf{u_1} + (x_3y_1 - x_1y_3)\mathbf{u_2} + (x_1y_2 - x_2y_1)\mathbf{u_3}$$

In other words, one can compute the cross product of $\mathbf{a}$ and $\mathbf{b}$ by computing the cross product of their coordinate vectors for any right handed orthonormal basis.

Theorem 12. (Lagrange's Identity.) Let $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}$. Then

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$$